

Unit Rates from Fractions and Mixed Numbers

Answer Key

Grades 6–7 Mathematics

Quick Answers

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|---|--|
| 1. $\frac{3}{2}$ mph | 8. $\frac{70}{3}$ |
| 2. $\frac{5}{2}$ km/h | 9. (a) price per kilogram at the first shop = 8, |
| 3. $\frac{7}{2}$ m/s | (b) price per kilogram at the second shop = $\frac{15}{2}$, |
| 4. $\frac{10}{3}$ ft/s | (c) comparison of the two prices = > |
| 5. $\frac{25}{3}$ | 10. 2 hours |
| 6. (a) rate of the first runner = $\frac{7}{6}$ mph, (b) rate | 11. $\frac{9}{4}$ |
| of the second runner = $\frac{5}{4}$ mph, (c) comparison | 12. (a) rate on the first leg = 7 mph, (b) rate |
| of the two rates = < | on the second leg = 5 mph, (c) rate for the |
| 7. $\frac{2}{5}$ hours | whole ride = $\frac{31}{5}$ mph, — |
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What is verified

18 of 19 answers machine-checked against SymPy, across 12 problems.

— marks an answer only you can judge — problem 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 6–7 Mathematics

6.RP.A.2, 6.RP.A.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*

Difficulty 1–5 of 5 across 19 tagged checks.

1. A runner covers $\frac{3}{4}$ mile in $\frac{1}{2}$ hour. Speed in miles per hour?

Solution: Miles per hour means miles divided by hours: $\frac{3}{4} \div \frac{1}{2} = \frac{3}{4} \times \frac{2}{1} = \frac{6}{4} = \frac{3}{2}$.

$\frac{3}{2}$ mph

The rate is larger than the distance, which is the sanity check: half an hour is less than one hour, so a full hour covers more ground.

2. A cyclist rides $\frac{5}{8}$ km in $\frac{1}{4}$ hour. Speed in kilometres per hour?

Solution: $\frac{5}{8} \div \frac{1}{4} = \frac{5}{8} \times \frac{4}{1} = \frac{20}{8} = \frac{5}{2}$.

$\frac{5}{2}$ km/h

3. A toy car travels $\frac{7}{8}$ metre in $\frac{1}{4}$ second. Speed in metres per second?

Solution: $\frac{7}{8} \div \frac{1}{4} = \frac{7}{8} \times \frac{4}{1} = \frac{28}{8} = \frac{7}{2}$.

$$\frac{7}{2} \text{ m/s}$$

4. A cart rolls $2\frac{1}{2}$ feet in $\frac{3}{4}$ second. Speed in feet per second?

Solution: Convert the mixed number first: $2\frac{1}{2} = \frac{5}{2}$.

Then divide: $\frac{5}{2} \div \frac{3}{4} = \frac{5}{2} \times \frac{4}{3} = \frac{20}{6} = \frac{10}{3}$.

$$\frac{10}{3} \text{ ft/s}$$

5. A printer produces $3\frac{1}{3}$ pages in $\frac{2}{5}$ minute. Pages per minute?

Solution: $3\frac{1}{3} = \frac{10}{3}$, and $\frac{10}{3} \div \frac{2}{5} = \frac{10}{3} \times \frac{5}{2} = \frac{50}{6} = \frac{25}{3}$.

$$\frac{25}{3}$$

That is $8\frac{1}{3}$ pages per minute.

6. First runner: $\frac{7}{8}$ mile in $\frac{3}{4}$ hour. Second runner: $\frac{5}{6}$ mile in $\frac{2}{3}$ hour. (a) First speed. (b) Second speed. (c) Compare.

Solution (a): $\frac{7}{8} \div \frac{3}{4} = \frac{7}{8} \times \frac{4}{3} = \frac{28}{24} = \frac{7}{6}$.

$$\frac{7}{6} \text{ mph}$$

(b) $\frac{5}{6} \div \frac{2}{3} = \frac{5}{6} \times \frac{3}{2} = \frac{15}{12} = \frac{5}{4}$.

$$\frac{5}{4} \text{ mph}$$

(c) Over the denominator 12: $\frac{7}{6} = \frac{14}{12}$ and $\frac{5}{4} = \frac{15}{12}$.

$$\frac{7}{6} < \frac{5}{4}$$

Neither runner's distance nor time is larger on its own in a way that settles this — only the rates do.

7. A pump fills $\frac{5}{6}$ of a tank in $\frac{1}{3}$ hour. How long for one whole tank?

Solution: The question asks for hours per tank, so divide the time by the fraction of the tank filled:

$$\frac{1}{3} \div \frac{5}{6} = \frac{1}{3} \times \frac{6}{5} = \frac{6}{15} = \frac{2}{5}$$

$$\frac{2}{5} \text{ hours}$$

Check the direction: filling more than the amount already done takes longer than $\frac{1}{3}$ hour, and $\frac{2}{5} > \frac{1}{3}$.

8. A car uses $\frac{2}{5}$ gallon to travel $9\frac{1}{3}$ miles. Miles per gallon?

Solution: $9\frac{1}{3} = \frac{28}{3}$, and miles per gallon is miles divided by gallons:

$$\frac{28}{3} \div \frac{2}{5} = \frac{28}{3} \times \frac{5}{2} = \frac{140}{6} = \frac{70}{3}$$

$\frac{70}{3}$

That is $23\frac{1}{3}$ miles per gallon.

- 9.** First shop: $\frac{3}{4}$ kg for 6 dollars. Second shop: $\frac{2}{3}$ kg for 5 dollars. (a) First price per kilogram. (b) Second price per kilogram. (c) Compare.

Solution (a): Dollars per kilogram is dollars divided by kilograms: $6 \div \frac{3}{4} = 6 \times \frac{4}{3} = \frac{24}{3}$.

8

(b) $5 \div \frac{2}{3} = 5 \times \frac{3}{2} = \frac{15}{2}$.

$\frac{15}{2}$

(c) $\frac{15}{2}$ is $7\frac{1}{2}$, which is less than 8.

$8 > \frac{15}{2}$

So the second shop is cheaper per kilogram, even though its total price is lower *and* its bag is smaller — the only fair comparison is per kilogram.

- 10.** A pump moves $\frac{7}{8}$ of a pool in $1\frac{3}{4}$ hours. How long for the whole pool?

Solution: First the rate: $1\frac{3}{4} = \frac{7}{4}$ hours, so the pump moves $\frac{7}{8} \div \frac{7}{4} = \frac{7}{8} \times \frac{4}{7} = \frac{1}{2}$ of a pool per hour.

Then the time for one whole pool: solving $\frac{1}{2}t = 1$ gives $t = 2$.

2 hours

- 11.** A recipe uses $1\frac{1}{2}$ cups of flour for $\frac{2}{3}$ of a batch. How many cups for a whole batch?

Solution: $1\frac{1}{2} = \frac{3}{2}$, and cups per batch is cups divided by batches:

$$\frac{3}{2} \div \frac{2}{3} = \frac{3}{2} \times \frac{3}{2} = \frac{9}{4}$$

$\frac{9}{4}$

That is $2\frac{1}{4}$ cups — more than $1\frac{1}{2}$, as it must be, since $\frac{2}{3}$ of a batch is less than a whole one.

- 12.** A cyclist rides $5\frac{1}{4}$ miles in $\frac{3}{4}$ hour, then $2\frac{1}{2}$ miles in $\frac{1}{2}$ hour. (a) First leg speed. (b) Second leg speed. (c) Whole-ride speed. (d) Why is (c) not the average of (a) and (b)?

Solution (a): $5\frac{1}{4} = \frac{21}{4}$, so $\frac{21}{4} \div \frac{3}{4} = \frac{21}{4} \times \frac{4}{3} = 7$.

7 mph

(b) $2\frac{1}{2} = \frac{5}{2}$, so $\frac{5}{2} \div \frac{1}{2} = \frac{5}{2} \times \frac{2}{1} = 5$.

5 mph

(c) Total distance $\frac{21}{4} + \frac{5}{2} = \frac{31}{4}$ miles; total time $\frac{3}{4} + \frac{1}{2} = \frac{5}{4}$ hours. Divide:

$$\frac{31}{4} \div \frac{5}{4} = \frac{31}{4} \times \frac{4}{5} = \frac{31}{5}$$

$\frac{31}{5}$ mph

(d) — **instructor-judged.** The plain average of 7 and 5 is 6, and the true answer is $\frac{31}{5}$, which is 6.2.

Look for a response built on total distance over total time: the two legs are not ridden for equal times, and the faster leg lasts $\frac{3}{4}$ of an hour against the slower leg's $\frac{1}{2}$, so the faster rate carries more weight. Averaging the two rates would only be right if the two times were equal.